Republic of Iraq
Ministry of Education and Scientific Research
University of Mosul
Computer Engineering Department

DESIGN AND IMPLEMENTATION OF A TACTILE STIMULATOR TO ENHANCE THE HUMAN SENSORY SYSTEM USING A PLOTTING-ON-THE-SKIN APPROACH

A graduation project is submitted to the Computer Engineering Department in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Engineering

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2024-2025

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DEDICATION

Dedicated to my family, whose unwavering support and encouragement have been my foundation; to my friends, who have shared in my journey and inspired me; and to all those who have imparted their knowledge and wisdom, guiding me along the way.

ACKNOWLEDGMENT

I want to express my heartfelt gratitude to everyone who has accompanied me throughout my academic life. Special thanks to my supervisor, Dr. Mazin Hashim Aziz, for his invaluable guidance and support during this research project. I am also grateful to my peers and colleagues for their collaboration and encouragement. This journey would not have been possible without the contributions of each individual who played a role in my development and success. Thank you all for your kindness and support.

ABSTRACT

This research presents the design, implementation, and evaluation of a novel tactile stimulator aimed at enhancing human sensory perception through a plotting-on-the-skin approach. Addressing limitations in existing sensory substitution devices, such as high cost, limited resolution, and extensive training requirements, this project developed a wireless wearable system centered around a modified Line-us robotic plotter. The system utilizes a custom Android application that processes visual input from a camera using YOLOv3-tiny object detection and a shape database to generate simplified drawing commands. These commands are transmitted wirelessly to the battery-modified Line-us plotter, which employs a feather brush to draw tactile patterns directly onto the user's skin. A user study involving 20 participants with mixed vision statuses and age groups was conducted over ten testing sessions to evaluate tactile shape recognition. Results demonstrated significant learning and improvement in recognition accuracy across participants, indicating the human sensory system's capacity to adapt to and interpret continuous tactile feedback delivered via plotting. Findings also highlighted the impact of shape complexity on recognition performance and provided valuable subjective feedback on the tactile interface. This work contributes a proof-of-concept for a potentially lower-cost and intuitive tactile sensory substitution method, offering a form of continuous tactile feedback. Future directions include leveraging artificial intelligence for real-time shape generation to overcome database limitations and exploring miniaturization possibilities through advancements like MEMS technology.

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1. INTRODUCTION

1.1. CONTEXT AND BACKGROUND

Visual impairments present significant challenges to individuals, impacting their ability to navigate their environments and access essential information. The World Health Organization estimates that approximately 2.2 billion people worldwide experience vision impairment or blindness, highlighting the urgent need for effective assistive technologies. [1] Sensory substitution techniques have emerged as a promising solution by converting visual information into other sensory modalities—such as tactile or auditory feedback—allowing visually impaired individuals to perceive their surroundings in new and meaningful ways.

Tactile sensory substitution methods leverage the brain's remarkable capacity for cross-modal processing, enabling users to interpret tactile sensations in ways analogous to visual perception. By translating visual stimuli into tactile feedback, these systems can enhance spatial awareness and improve mobility for users. [2] Technologies such as vibrating devices, pressure sensors, and tactile displays have been developed to convey visual information through touch. However, the effectiveness of these systems often relies on user training and adaptability, which can vary significantly among individuals.

1.2. PROBLEM STATEMENT

Despite advancements in sensory substitution technologies, several limitations persist. Many existing devices require extensive training and adaptation periods, which can limit their accessibility and practicality for everyday use. [3] Additionally, the effectiveness of these systems in dynamic, real-world environments is frequently untested, raising questions about their reliability and usability across various contexts. There is a pressing need for innovative tactile-based solutions that are intuitive, easy to use, and capable of providing meaningful feedback to users.

This project aims to address these gaps by exploring a tactile-based system that utilizes a Line-us Plotter to create continuous tactile drawings on the skin. This method seeks to provide a straightforward and interpretable means of conveying visual information, potentially enhancing user experience and effectiveness in sensory substitution devices.

1.3. OBJECTIVE OF THE RESEARCH

The primary objective of this research is to develop a tactile-based system that translates **visual information** into tactile feedback using a plotter on the skin. By integrating a Line-us Plotter with a visual processing pipeline based on **object detection and a shape database**, the system will draw simplified shapes and patterns directly onto the skin, allowing users to perceive and interpret information from their surroundings through touch. This innovative approach aspires to bridge the gap between visual and tactile perception, ultimately improving the spatial awareness of visually impaired individuals and exploring tactile perception in a mixed user group.

1.4. RESEARCH QUESTIONS

This research will address several key questions:

1. How can tactile feedback improve the spatial awareness of blind individuals?
2. What are the limitations and challenges of implementing such a tactile-based system?
3. In what ways can the use of a plotter enhance the effectiveness of tactile sensory substitution?

1.5. SIGNIFICANCE OF THE STUDY

The findings from this study have the potential to make substantial contributions to the field of assistive technology for the visually impaired. By developing a more intuitive and accessible tactile feedback system, we aim to empower individuals with visual impairments to navigate their environments more effectively and access information that was previously out of reach. [4] This research could pave the way for future innovations in sensory substitution technologies, ultimately enhancing the quality of life for millions of individuals worldwide. Specifically, this study presents a novel **plotter-on-the-skin approach** for tactile sensory substitution, offering a method for delivering **continuous tactile feedback** that differs from traditional discrete tactile displays. By leveraging a **low-cost off-the-shelf plotter** and a smartphone-based system, it explores a more accessible pathway for developing such technologies.

Through the **empirical evaluation** involving a mixed group of participants, this research provides valuable insights into the human capacity for **learning and interpreting tactile patterns** drawn on the skin,

contributing data on the effectiveness and factors influencing performance with this specific tactile modality.

2. LITERATURE REVIEW

2.1. OVERVIEW OF SENSORY SUBSTITUTION

Sensory substitution is a transformative approach that seeks to compensate for a lost sense by converting information typically accessed through one sensory modality into another. This field has significant implications for individuals with visual impairments, offering alternative pathways to perceive their environment. The foundational work by [5] demonstrated that tactile stimulation could convey visual information, thereby activating the visual cortex in blind individuals. This pioneering research laid the groundwork for subsequent studies exploring how other senses can compensate for vision loss.

Recent advancements have further expanded the scope of sensory substitution, integrating modern technology with neuroscience. Devices like the BrainPort (**Figure 2.1**) and vOICe have shown that non-visual modalities can effectively transmit spatial information, enabling users to interpret their surroundings through taste or sound. [5], [6] These innovations underscore the potential of sensory substitution to enhance the quality of life for visually impaired individuals. The integration of machine learning and artificial intelligence has also opened new avenues for enhancing the effectiveness of these systems, allowing for more personalized and adaptive sensory experiences. [7]

Figure 2.1 | The BrainPort Vision Device is a sensory substitution system that down-samples video from a glasses-mounted frame camera to a 400-pixel (20 × 20) electro-tactile array.

Beyond aiding those with visual impairments, sensory substitution offers profound insights into human perception, cognition, and neuroplasticity, [8] potentially impacting neurorehabilitation. [9] It challenges traditional views that treat senses as distinct channels by demonstrating the brain's capacity to effectively integrate and interpret information from different sensory modalities. This has implications for our understanding of how we construct our perceptual experience and how we might design technologies that enhance or augment these experiences.

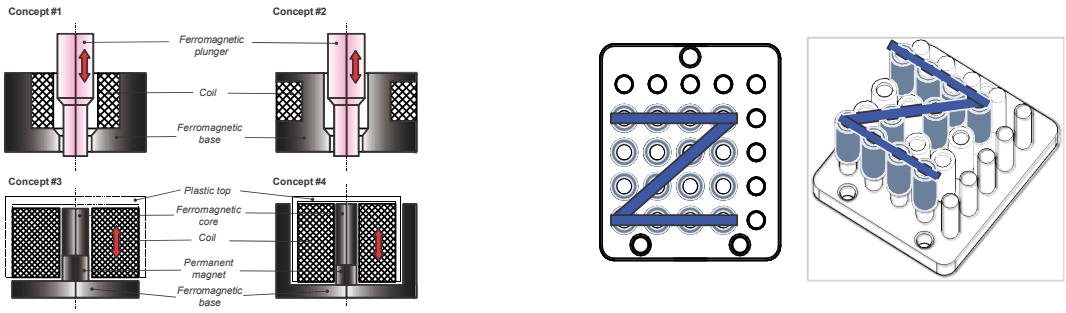
2.2. TACTILE FEEDBACK SYSTEMS

Tactile feedback systems have evolved significantly, employing various mechanisms to relay spatial and environmental data. These systems can be broadly categorized into electromagnetic, thermal, and mechanical, etc. Electromagnetic systems, for instance, use magnetic fields to generate tactile sensations, while thermal systems rely on temperature changes. Mechanical systems, such as those utilizing micro-actuator arrays (**Figure**

2.2), physically manipulate the skin to create perceivable patterns. [10] A significant challenge for many such tactile displays, particularly those relying on discrete actuator arrays, is their limited spatial resolution, which can hinder the conveyance of fine details or complex patterns.

As an alternative to discrete actuator arrays, this research explores the concept of creating continuous tactile patterns or 'drawing' directly on the skin using a mechanical plotter. This approach may offer a different method for conveying spatial information with a resolution potentially limited by the contact point and skin sensitivity.

The effectiveness of tactile displays hinges on their ability to deliver clear and distinct signals that users can interpret without ambiguity. Research by [11] explored open-source portable haptic displays, emphasizing the importance of integrating visual-force-tactile feedback for enhanced user experience. Additionally, studies by [12] and [4] have contributed to understanding how different tactile display technologies can be optimized for sensory substitution.



(a) (b)

Figure 2.2 | Micro-actuator arrays. (a) Examined concepts of electromagnetic actuators. (b) Coils forming the letter Z.

It is important to note that tactile sensations vary across different areas of the human skin (**Figure 2.3**), which is the largest organ in the human body. This variability can be attributed to differences in skin structure, density of

mechanoreceptors, and sensitivity levels across various regions. For example, areas such as the fingertips and palms are densely packed with

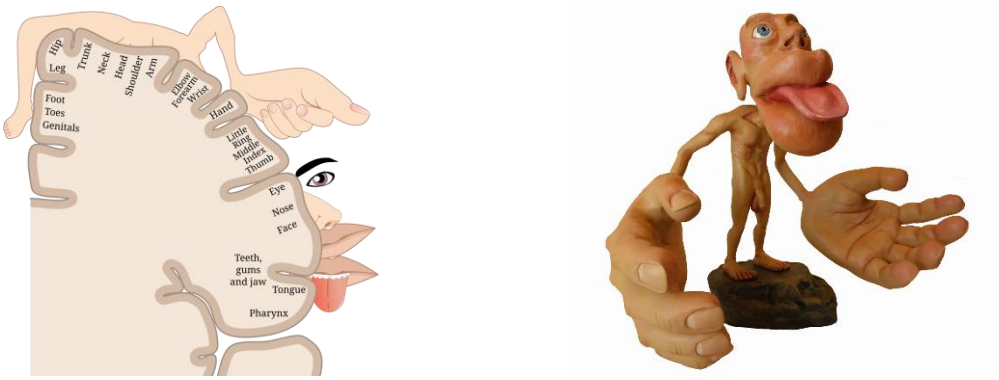


Figure 2.3 | Sensory Homunculus. It shows how high the tactile sensation is across different areas of the human skin.

sensory receptors, making them more sensitive to touch compared to less sensitive areas like the forearm or back. This characteristic can be leveraged in tactile feedback systems, allowing for a more effective interpretation of visual information through tactile sensations applied to the skin. [1]

Further exploration into the design and application of tactile displays reveals a growing interest in wearable technologies. These include gloves and vests equipped with sensors and actuators that provide tactile feedback in response to external stimuli. [12] Such innovations aim to offer more intuitive and immersive experiences, bridging the gap between virtual and physical environments. In the context of virtual reality (VR) and augmented reality (AR), tactile feedback enhances user interaction by providing a sense of touch, which can significantly improve the realism and engagement of VR/AR experiences. [4]

Moreover, the development of advanced materials and actuator technologies has led to the creation of more sophisticated tactile feedback systems. For example, thermo-activated tactile micro-actuators have been developed to provide dynamic and adaptive feedback, enhancing the user's

sensory experience. [13] These advancements highlight the potential for tactile feedback systems to become more integrated into everyday technology, offering new ways to interact with digital content.

Research also indicates that tactile feedback can play a significant role in enhancing motor skills and coordination. Studies have shown that providing users with tactile feedback during tasks can improve their performance and learning outcomes. [14] This suggests that tactile feedback systems could be beneficial not only for individuals with sensory impairments but also for enhancing skill acquisition in various domains, including sports, rehabilitation, and education.

2.3. NEUROLOGICAL INSIGHTS

Cross-modal plasticity is a key neurological phenomenon underpinning sensory substitution. It refers to the brain's ability to reorganize itself by forming new neural connections, particularly when a sensory modality is absent or impaired. Studies by [8] have shown that the visual cortex can process tactile and auditory stimuli in blind individuals,

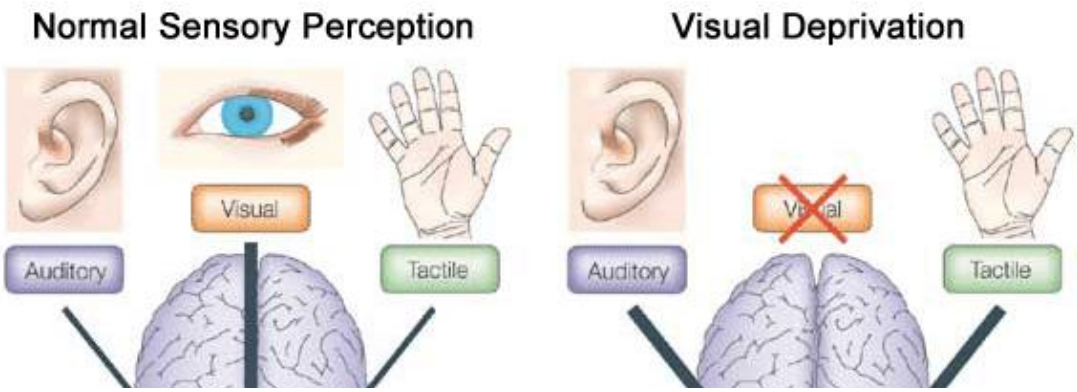


Figure 2.4 | After visual deprivation, neuroplastic changes occur such that the visual cortex is recruited to process sensory information from other senses (illustrated by larger arrows for touch and hearing).

demonstrating the brain's remarkable adaptability[15] (**Figure 2.4**) – a principle leveraged by sensory substitution technologies.

The capacity for cross-modal plasticity is influenced by various factors, including the age of the individual and the duration of sensory deprivation. Research indicates that younger individuals exhibit greater neuroplasticity (**Figure 2.5**), allowing them to adapt more readily to alternative sensory inputs. [2], [16] This suggests that early intervention and training can significantly enhance the brain's ability to process non-visual stimuli, thereby improving the effectiveness of sensory substitution devices.

Neuroimaging studies have further elucidated the mechanisms underlying cross-modal processing. For instance, findings indicate that the visual cortex not only responds to tactile stimuli but can also integrate auditory information, demonstrating a high degree of interconnectivity among sensory areas of the brain. [17] This interconnectedness underscores the potential for developing sensory substitution devices that incorporate multiple modalities, providing users with a richer sensory experience.

Additionally, the study of cross-modal plasticity offers valuable insights into the fundamental workings of the brain, particularly regarding sensory integration and perception. Understanding how the brain adapts to

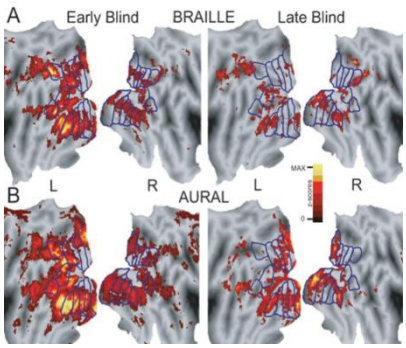


Figure 2.5 | Two-dimensional, flattened views of z-score statistical parameter maps for visual cortex BOLD responses in early and late blind subjects

new sensory inputs can inform the design of innovative therapies for individuals with sensory processing disorders or other neurological conditions. [9] Moreover, these insights can guide the development of strategies to enhance cognitive functions and sensory integration in both clinical and non-clinical populations.

Ultimately, understanding cross-modal plasticity is crucial for designing sensory substitution devices responsive to individual needs. By capitalizing on the brain's inherent ability to reorganize and integrate sensory information, researchers can create more effective for individuals with visual impairments.

2.4. EXISTING RESEARCH AND TECHNOLOGIES

Current sensory substitution devices vary in complexity and application. The BrainPort device (previously mentioned), for instance, translates visual information into electrical signals delivered to the tongue, allowing users to "see" through taste. [5] Similarly, the vOICe system converts visual images into soundscapes, which users learn to interpret as visual scenes. [6]

However, these technologies often face limitations, such as steep learning curves and high costs, which can hinder widespread adoption. Recent advancements in wearable tactile technology, previously discussed for their immersive potential (see Section 2.2), also aim to address these challenges by offering more intuitive and affordable solutions. [7]

Emerging research focuses on improving the accessibility and usability of sensory substitution devices. For example, [3] explores the potential of low-cost, open-source solutions that leverage existing consumer electronics to provide sensory feedback. This approach aims to democratize

access to sensory substitution technologies, making them available to a broader audience.

In addition to improving accessibility, there is a growing emphasis on enhancing the user experience of sensory substitution devices. This includes developing more intuitive interfaces and incorporating multimodal feedback to provide a richer sensory experience. By combining tactile, auditory, and visual feedback, these devices can offer more comprehensive support for users, allowing them to better navigate and interact with their environment. [4]

There is growing interest in using these technologies to augment human capabilities, providing enhanced sensory experiences for all users. This includes applications in entertainment, education, and professional settings, where sensory substitution can offer new ways to engage with content and perform tasks.

Additionally, recent studies have explored innovative interfaces for non-visual graphical information processing. [18] introduced a touch-based vibro-audio interface that enables users to learn graphical information through tactile and auditory cues, demonstrating the effectiveness of multimodal sensory feedback in enhancing understanding and interaction with graphical data. Furthermore, [19] work on auditory image representations showcases the potential of auditory feedback in conveying complex visual information, further emphasizing the importance of integrating multiple sensory modalities in sensory substitution systems.

2.5. GAPS IN CURRENT RESEARCH

Despite the progress in sensory substitution, several gaps remain. Many existing devices require extensive training and adaptation periods,

limiting their accessibility. Moreover, the effectiveness of these systems in dynamic, real-world environments is often untested. Our project, which utilizes a plotter-based tactile system, addresses these gaps. By drawing directly on the skin, it aims to eliminate the resolution problem inherent in many digital tactile displays that rely on discrete actuators. Furthermore, this method provides a continuous representation of tactile information, offering a potentially more intuitive way of conveying visual data through simple, interpretable shapes.

This novel approach not only leverages the brain's capacity for cross-modal processing but also aims to provide a more immediate and natural form of interaction for users. By focusing on the tactile modality, our research contributes to a growing body of work seeking to provide more effective and accessible technological solutions for visually impaired individuals. The use of a mini plotter to convey visual data through tactile feedback represents a significant advancement in the field, promising to make sensory substitution more accessible and effective.

The plotter-based approach offers several advantages over traditional sensory substitution methods. Directly drawing shapes on the skin provides a more tangible and immediate form of feedback, reducing the cognitive load on users and allowing for faster interpretation of sensory information. This method also has the potential to be more cost-effective and easier to implement than other tactile systems, making it an attractive option for widespread adoption.

Furthermore, the plotter-based system's adaptability allows it to be customized for individual users, tailoring the feedback to their specific needs and preferences. This personalization can enhance the effectiveness of the system, ensuring that it meets the unique requirements of each user.

2.6. FUTURE DIRECTIONS IN TACTILE SENSORY SUBSTITUTION

The future of tactile sensory substitution lies in the integration of advanced technologies and interdisciplinary research. As computational power and sensor technology continue to evolve, there is potential for creating more sophisticated and responsive tactile displays. Research into materials science, for instance, could lead to the development of more sensitive and flexible actuators that conform to the body's contours, providing more natural and comfortable feedback. [13] (**Figure 2.6**)

A significant area of advancement in actuator technology involves Micro-Electro-Mechanical Systems (MEMS). MEMS technology offers the potential to create tactile displays with significantly smaller actuator sizes, enabling much higher resolution and more detailed tactile feedback. [20] For instance, research has demonstrated scalable, MEMS-enabled vibrational tactile actuators that can be arrayed to form high-resolution tactile displays, paving the way for more sophisticated methods of conveying complex spatial information. [21] This miniaturization and increased density of tactors are crucial for developing future tactile interfaces capable of rendering fine textures and intricate patterns.

Further advancements in machine learning could significantly enhance the ongoing efforts towards personalization of sensory substitution devices, allowing them to more dynamically adapt to individual user preferences and needs. This adaptability could significantly improve user satisfaction and the overall effectiveness of these technologies. [14]

Collaborative efforts between engineers, neuroscientists, and clinicians will be crucial in advancing the field of tactile sensory substitution. By combining insights from these diverse disciplines, it is possible to create comprehensive solutions that address the complex needs of visually impaired individuals, ultimately improving their independence.

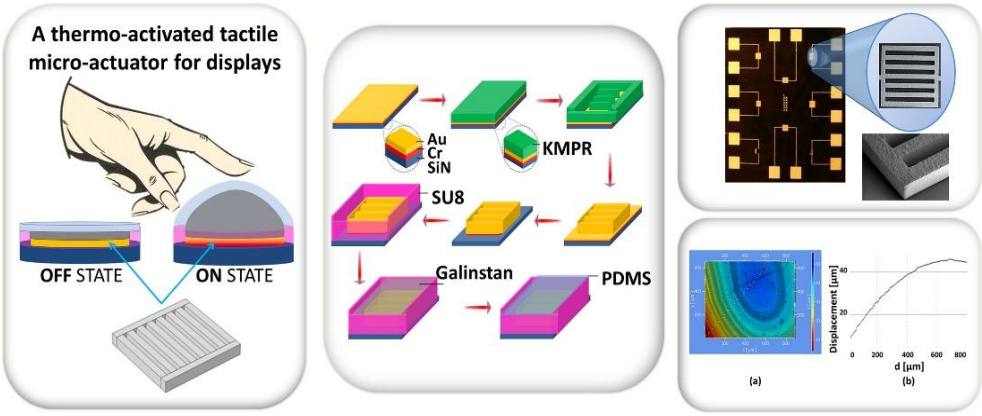


Figure 2.6 | A thermo-activated tactile micro-actuator for displays

Building on their current applications in areas like VR/AR (Section 2.2), these technologies have broader potential to enhance human-computer interaction and augment reality experiences, providing users with new ways to perceive and interact with digital content. As the field continues to evolve, it will be important to consider the ethical implications of sensory substitution technologies, ensuring that they are developed and implemented in ways that respect user autonomy and privacy.

The potential for tactile sensory substitution to transform various industries is vast. In the realm of education, for example, tactile feedback can provide students with new ways to engage with learning materials, making abstract concepts more tangible and accessible. In healthcare, these technologies can assist in rehabilitation and therapy, offering patients new tools to regain lost functions.

3. METHODOLOGY

3.1. PROJECT DESIGN

This subsection describes the final system architecture and the rationale behind the chosen approach, including a brief overview of alternative designs explored.

3.1.1. Overview of the System

The developed system is a wireless, wearable tactile stimulator designed to enhance human sensory perception by translating visual information into dynamic tactile feedback on the skin. The core of the system utilizes a modified Line-us robotic plotter to create continuous line drawings directly onto the user's forearm or hand (optionally can be placed on other body parts). Visual input is captured via a camera (either the built-in camera of an Android smartphone or an external wireless webcam), processed by a custom Android application, and converted into a sequence of plotting commands. These commands are then transmitted wirelessly to the Line-us device for execution. This approach aims to leverage the skin's tactile sensitivity to provide users with a novel method for perceiving simplified representations of their environment. Additionally, a separate prototype of a thermal tactile display was developed and explored to investigate alternative

methods of conveying sensory information through temperature changes, serving as a comparative element to the primary plotter-based system.

3.1.2. System Architecture

The system architecture is designed around a modular structure, facilitating the flow of information from visual input to tactile output. At its core is a custom-developed **Android application**, which serves as the central processing and control unit. This application manages the connection to the camera input and the Line-us plotter, processes the visual data, and generates the tactile drawing commands. (**Figure 3.1**)

The architecture can be broadly divided into the following interconnected modules:

1. **Input Module:** This module captures visual information from the environment. The system is designed to be flexible, supporting input from either the **built-in camera of the Android smartphone** running the application or an **external wireless webcam**.
2. **Processing Module:** Residing within the Android application, this module receives the camera feed and performs the necessary analysis. It incorporates an **Object Detection Sub-module** utilizing the YOLOv3-tiny model to identify objects within the frame. Based on the detected objects, a target object is selected (e.g., the object closest to the center of the frame). The size of the selected object is used to determine a dynamic scaling factor.
3. **Shape Mapping Module:** Following object detection, the class label of the selected object and the calculated scale factor are used to query a pre-defined **Shape Database**. This database stores simplified drawing commands (G-code sequences) corresponding to various

object classes. The module retrieves the appropriate scaled command sequence for the detected object.

4. **Communication Module:** This module handles the wireless communication link between the Android device and the Line-us plotter. It establishes and maintains a connection via Wi-Fi and transmits the generated G-code commands to the plotter.
5. **Tactile Output Module:** The **Line-us Plotter**, modified for standalone operation with an integrated battery and fitted with a suitable pen head for skin contact, receives the G-code commands. It executes these commands by moving its arm and controlling the pen state (up/down) to draw the tactile pattern on the user's skin.
6. **User Interface Module:** Integrated within the Android application, the user interface provides a visual display of the live camera feed (overlaid with detection bounding boxes), a preview of the drawing commands to be sent to the plotter, and status information about the system's operation (e.g., detection status, drawing progress, connection status). It also includes a settings interface for configuring system parameters.

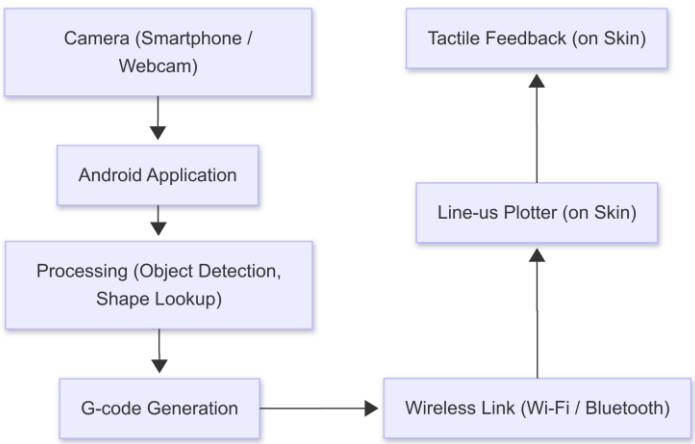


Figure 3.1| Flowchart illustrating the architecture of the plotter-based tactile stimulator system.

The system operates in real-time, with the processing and communication modules working in tandem to provide responsive tactile feedback based on the dynamic visual input. The drawing execution on the Line-us is managed in a separate thread within the application to ensure the user interface remains responsive.

For the separate exploration of thermal tactile feedback, a **Thermal Tactile Display Prototype** was also developed. This prototype consists of a 5x4 resistor matrix controlled by an ESP32 microcontroller. While not fully integrated into the main visual-to-tactile pipeline in this phase, it represents an investigation into alternative tactile modalities for future development.

3.1.3. Design Evolution and Rationale

The final architecture of the tactile stimulator system was the result of an iterative design process that involved exploring and evaluating multiple approaches for translating visual input into tactile drawing commands. Initially, three primary methods were investigated to determine the most effective strategy for generating clear and meaningful tactile feedback on the skin. (See **Figure 3.2**)

The first approach explored relied heavily on **traditional image processing techniques**. This method involved applying steps such as edge detection (Canny) and contour finding to the raw camera feed. The detected contours were then subjected to simplification algorithms (such as Ramer-Douglas-Peucker) and a grouping logic based on centrality and proximity to focus on the main subject in the frame. While this approach offered the potential to draw the actual perceived outlines of objects, it presented significant challenges. It proved highly sensitive to variations in lighting conditions, background clutter, and image noise, often resulting in

fragmented or inaccurate contour detection. Reliably isolating and grouping contours belonging to a single object in dynamic, real-world environments was particularly difficult.

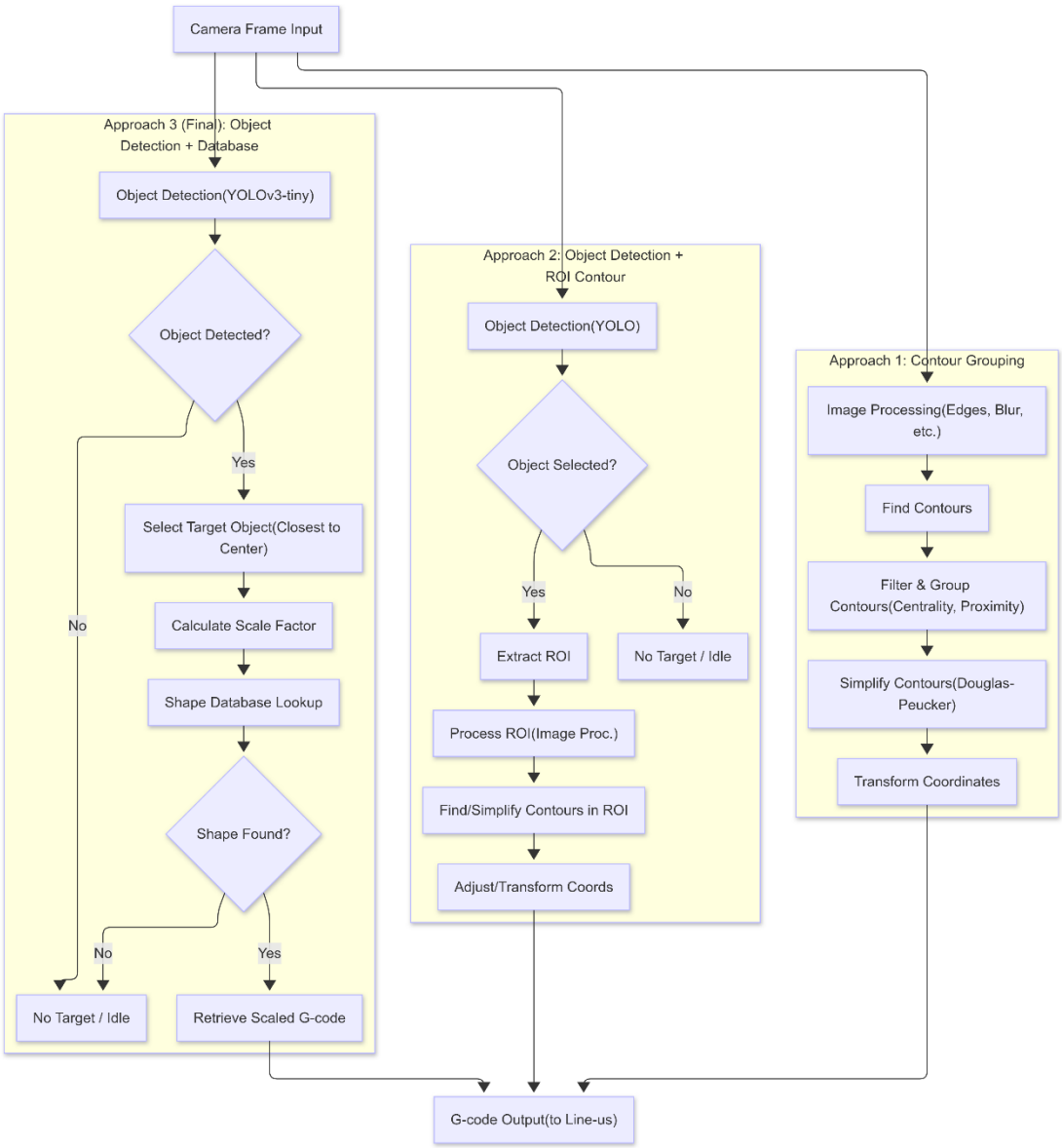


Figure 3.2 | Is the flowchart summarizing the three approaches.

A second approach aimed to mitigate the clutter sensitivity by combining **object detection with region-of-interest (ROI) contour processing**. In this method, an object detection model (initially YOLOv3-tiny, later YOLOv3) was used first to identify objects and define a bounding

box around the primary target (e.g., the object closest to the frame center). Image processing techniques (blurring, Canny edge detection, morphological operations, contour finding, and simplification) were then applied *only* within this defined ROI. This significantly improved robustness against background clutter compared to the first approach, as processing was confined to the detected object's area. However, the quality and consistency of the resulting tactile drawing still depended heavily on the clarity of the object's outline within the potentially low-resolution or partially obscured ROI from the camera feed. Tuning the image processing parameters to work consistently across different objects and conditions within the ROI remained challenging.

Based on the limitations observed with the first two methods, the third approach, **Object Detection coupled with Shape Database Mapping**, was adopted for the final system implementation. This strategy utilizes the object detection model (YOLOv3-tiny) to identify the class of the target object. Instead of attempting to process the object's live contour, the system retrieves a pre-defined, simplified drawing sequence (G-code) for that specific object class from a curated database. The size of the detected object's bounding box is used to dynamically scale this pre-defined shape. This approach offered several key advantages that addressed the shortcomings of the previous methods:

- **Speed and Performance:** By replacing complex, per-frame contour extraction with a fast object detection lookup, the processing pipeline became significantly faster and more efficient.
- **Robustness:** Relying on object detection to identify known classes made the system much more robust to background clutter and variations in object appearance or partial occlusion, as the drawing is

generated from a reliable database entry rather than potentially noisy live contours.

- **Consistency and Simplicity:** The output tactile patterns are consistent pre-defined shapes, making them potentially easier for users to learn and recognize compared to variable, simplified contours derived from live images.
- **Focus:** The system reliably focuses on drawing a representation of the identified target object, ignoring irrelevant details or background elements.

While this approach is limited to drawing objects present in the shape database, the benefits in terms of speed, robustness, and consistency made it the most suitable method for developing a functional and evaluable prototype for tactile sensory substitution within the scope of this project.

3.1.4. Wearable Design

A critical aspect of the project design was the development of a wearable configuration to allow the Line-us plotter to deliver tactile feedback directly onto the user's skin. The primary area targeted for tactile stimulation was the forearm or hand, selected for its accessibility and relatively high tactile sensitivity.

The standard Line-us plotter, typically designed for drawing on paper while connected to a power source, was modified to enhance its portability and autonomy. A small battery was integrated with the Line-us device, enabling it to operate wirelessly without being tethered to a power outlet. This modification significantly improved the user's freedom of movement and the practicality of the system as a wearable device.

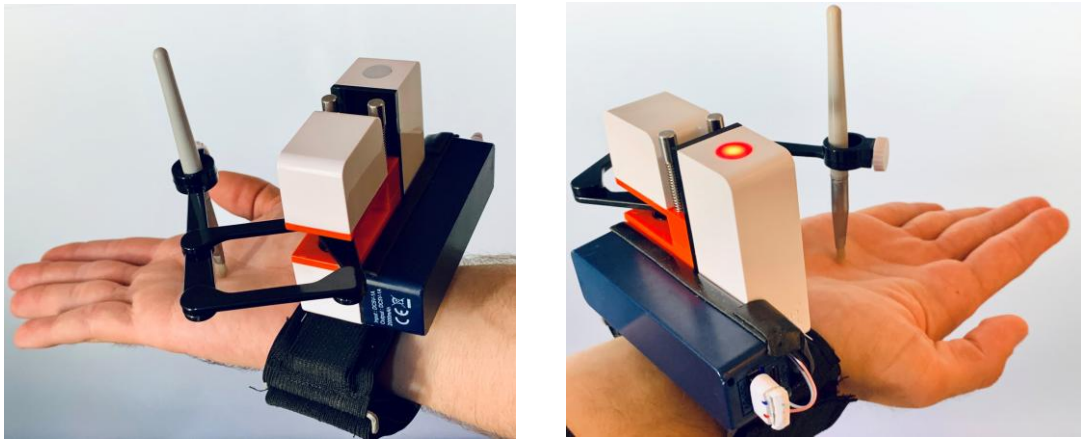


Figure 3.3 | Wearable Design.

A mounting mechanism was designed to securely attach the modified Line-us plotter to the user's arm or hand. This mount ensured that the plotter remained in a stable position relative to the skin surface, allowing for consistent and accurate drawing of tactile patterns. The design prioritized user comfort to facilitate extended use during training and testing sessions.

The tactile interface with the skin was achieved by fitting the Line-us arm with various pen heads designed for safe and effective skin contact (as detailed in Section 3.2.2). The plotter's ability to control the Z-axis (pen up/down) allowed for precise control over when and how the pen head made contact with the skin, creating the intended tactile sensations.

3.2. HARDWARE COMPONENTS

This section describes the physical components utilized in the design and implementation of the tactile stimulator system, including the main plotter-based hardware and the separate thermal tactile display prototype.

3.2.1. Line-us Plotter

The core hardware component providing tactile output is the **Line-us robotic plotter**. (**Figure 3.4**) This device is controlled via G-code commands transmitted wirelessly (as detailed in Section 3.3.3), allowing precise manipulation of its arm in three dimensions. The Line-us operates within a defined workspace, typically corresponding to a physical area of approximately 100mm x 100mm. This physical space is mapped to a digital coordinate system, commonly ranging from 0 to 2000 units in both the X and Y axes (with 100 units equivalent to 5mm). The Z-axis controls the vertical

Figure 3.4 | Line-us plotter.

position, differentiating between pen-up (PEN_UP_Z) and pen-down (PEN_DOWN_Z) states to control contact with the drawing surface.

For this project's application as a wearable tactile stimulator, the standard Line-us was modified to operate autonomously by integrating a small battery, eliminating the need for a wired power connection during use. Furthermore, while the Line-us has a full drawable area, a specific sub-region was selected for consistently plotting tactile shapes on the user's skin. This chosen application drawing area spans from coordinates (700, -900) to (1700, 100) within the Line-us's coordinate system. The system's software pipeline is designed to scale and center the tactile representation of the

detected object within this designated area on the skin, regardless of the object's position in the camera frame.

Information regarding the Line-us's technical specifications, communication protocol, coordinate system, and programming interface was referenced from the official Line-us programming documentation. [22]

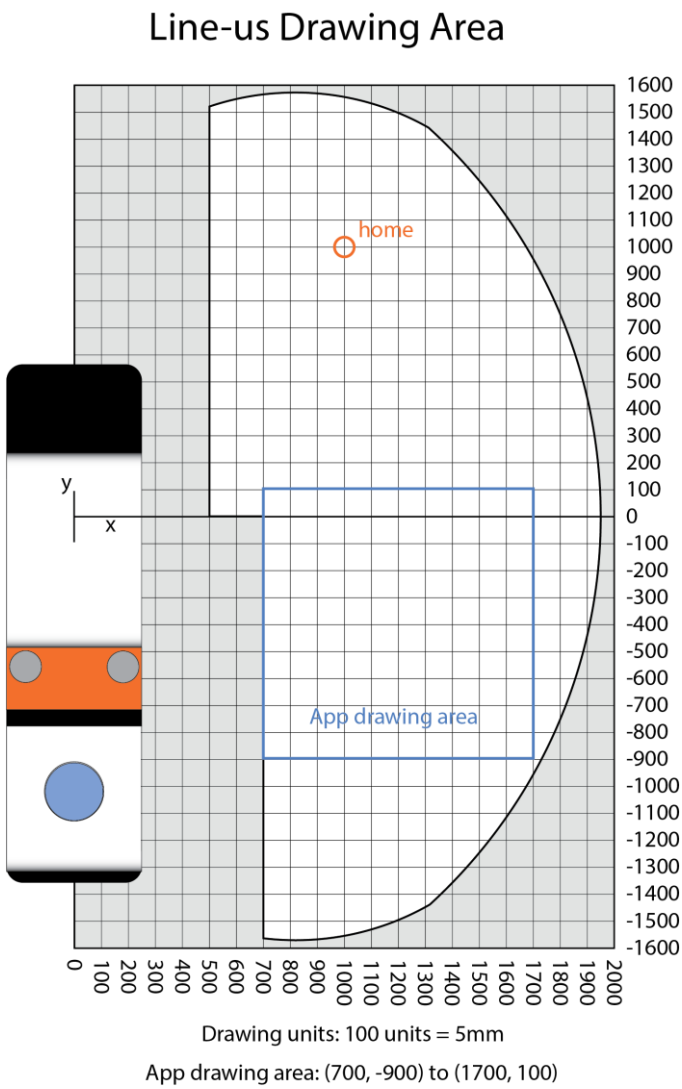


Figure 3.5 | Illustration of the full drawable area of the Line-us plotter and highlight of the specific sub-region utilized for tactile drawing in this project.

3.2.2. Tactile Sensing Pen-Heads

Various pen heads were explored to determine the most suitable for delivering tactile feedback directly onto the skin. Initial testing included a standard stylus pen, a pen with a thin nylon piece (used with disposable plastic gloves), a feather brush, and a head with a small moving ball. Based on these experiments, the feather brush was ultimately selected for the main user study due to several advantages. The brush demonstrated robustness against minor obstacles and variations in hand geometry during movement. Its soft bristles provided a gentle and comfortable tactile sensation on the skin, while still offering good sensitivity for shape recognition. (**Figure 3.6**)

Figure 3.6 | Testing different tools on the skin and evaluating them.

3.2.3. Microcontroller

The primary computational power for the main plotter-based system is provided by the **Android smartphone** running the custom-developed Kivy application. This device handles the camera input, executes the object

detection and shape lookup algorithms, manages the communication with the Line-us plotter, and renders the user interface.

3.2.4. Camera

Visual input for the system is acquired through a camera. The design offers flexibility in the input source, allowing the use of either the **built-in camera** of the Android smartphone hosting the application or an **external wireless webcam**. During development and testing, an IP Webcam setup utilizing a smartphone was employed as the external camera source.

3.2.5. User Interface

The primary user interface is the **Android smartphone** running the custom Kivy application (detailed in Section 3.3.1), serving as the display and touch input device. This provides visual feedback and control for sighted users or during setup. For enhanced accessibility, particularly for visually impaired users, the system design allows for the potential integration of an **optional hardware control piece**, such as a remote, to provide tactile or audio feedback for system control and navigation, reducing reliance on the visual screen interface.

3.2.6. Thermal Tactile Display Prototype Hardware

Aa a separate exploratory component, a prototype thermal tactile display was constructed. The hardware for this prototype consists of a **5x4 matrix of 100-ohm resistors** mounted on a surface. Control is facilitated by an **ESP32 microcontroller**, utilizing **ULN drivers** to manage power to the resistors. A **cooling fan** was also included in the prototype design, positioned

above the resistor matrix to aid in faster temperature dissipation, although its integration into the control logic was limited in this phase. This prototype was developed to investigate the feasibility of thermal feedback as an alternative tactile modality. (See below **Figure 3.7**)

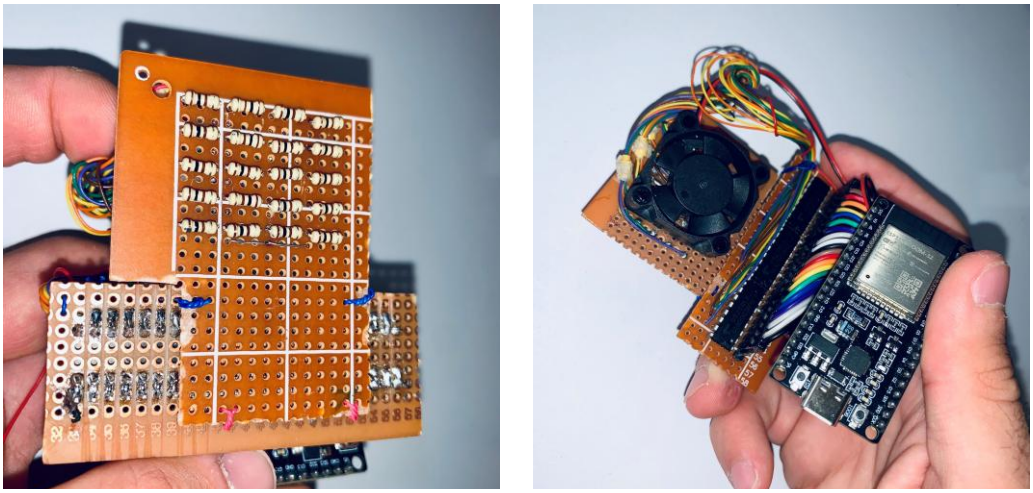


Figure 3.7 | Thermal Tactile Display.

3.3. SOFTWARE COMPONENTS

This section details the software developed and utilized for the tactile stimulator system. The software is responsible for acquiring and processing visual data, implementing the object detection and shape mapping pipeline, controlling the Line-us plotter, managing the user interface, and handling communication.

3.3.1. Android Application (Kivy)

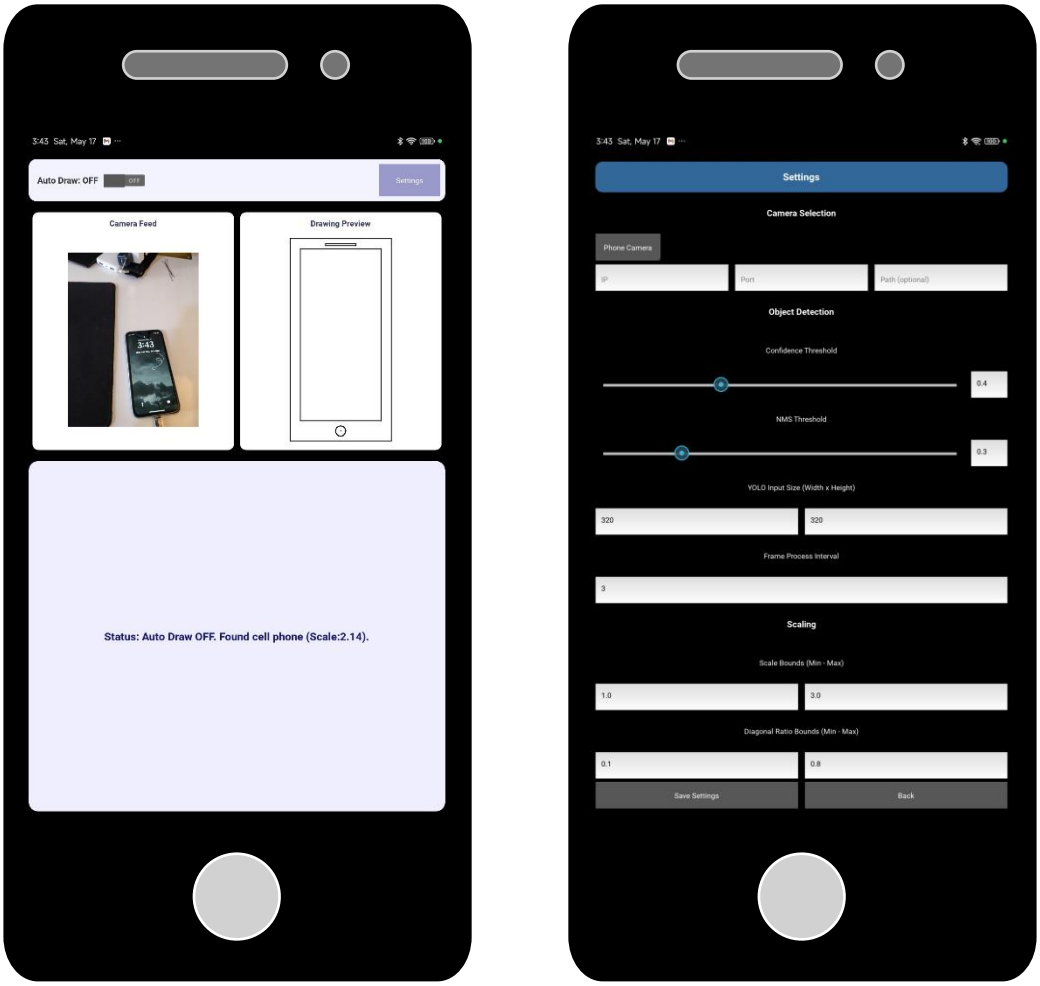
The central software component of the system is a custom application developed for the Android platform using the **Kivy framework**. This application serves as the primary interface and control hub, integrating the various hardware and software modules. Its main responsibilities include acquiring the camera feed, executing the visual processing pipeline,

managing the wireless communication with the Line-us plotter, and providing a user interface for monitoring and configuration. (**Figure 3.8**)

Key functionalities of the Android application include:

- **Camera Management:** Handling the interface with the selected camera source, whether it's the built-in smartphone camera or an external webcam stream.
- **Visual Processing Pipeline Execution:** Running the object detection (YOLOv3-tiny) and shape lookup logic on the captured frames.
- **Line-us Communication Control:** Initiating and managing the TCP/IP socket connection to the Line-us plotter and sending the generated G-code commands. This communication is handled in a separate thread to maintain application responsiveness.
- **Visual Feedback:** Displaying the live camera feed, often overlaid with object detection bounding boxes, and rendering a real-time preview of the tactile pattern being generated for the Line-us.
- **Status Monitoring:** Presenting status messages to the user regarding system initialization, object detection status, Line-us connection, and drawing progress.
- **Settings Configuration:** Providing an interface for users to adjust various system parameters, such as camera source, object detection thresholds, scaling parameters, and Line-us connection details.

The Kivy framework was chosen for its ability to facilitate cross-platform development, enabling the application to run on a standard Android smartphone, thus leveraging readily available mobile hardware.



(a) Main Screen (b) Settings Screen

Figure 3.8 | The Android Application (Kivy) preview.

3.3.2. Object Detection and Shape Lookup Pipeline

The system's ability to translate visual information into meaningful tactile feedback relies on a software pipeline implemented within the Android application. This pipeline processes the camera feed to identify

objects and retrieve corresponding drawing instructions from a pre-defined database. (See below **Figure 3.9**)

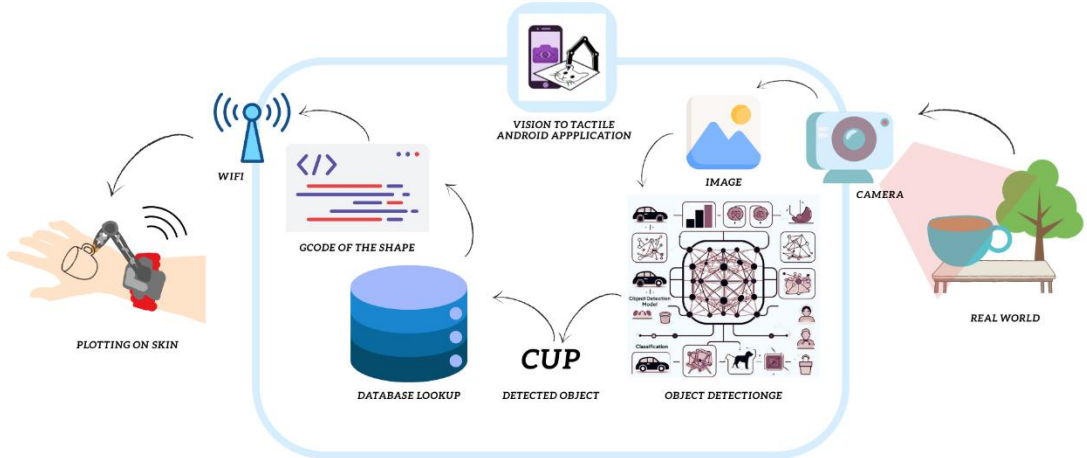


Figure 3.9 | Simplifies the approach and what happens inside the application.

The initial step involves **Object Detection**, where the system utilizes **OpenCV's Deep Neural Network (DNN) module** loaded with the **YOLOv3-tiny model**. This compact convolutional neural network, along with its pre-trained weights (yolov3-tiny.weights) and configuration file (yolov3-tiny.cfg), is used to analyze frames from the camera input. The model is trained on the COCO dataset, enabling it to detect a wide variety of common objects defined in the coco.names file. Key parameters, including the confidence threshold (CONFIDENCE_THRESHOLD) and Non-Maximum Suppression threshold (NMS_THRESHOLD), are applied to filter detections and identify the most probable objects and their bounding boxes within the frame.

Following object detection, the system implements a **Target Selection** mechanism. From the list of detected objects, the object whose bounding box centroid is geometrically closest to the center of the camera frame is selected as the primary target for tactile representation. This

approach ensures that the system focuses on the most prominent or central object in the user's field of view.

For the selected target object, a **Dynamic Scaling** factor is calculated. This factor is determined based on the relative size of the object's bounding box compared to the overall dimensions of the camera frame. Specifically, the ratio of the object's bounding box diagonal to the image diagonal is mapped to a predefined range of scale factors (MIN_SCALE_FACTOR to MAX_SCALE_FACTOR). This allows the tactile drawing size to adapt dynamically based on the perceived distance or size of the object in the camera feed.

Finally, the system performs **Shape Database Mapping**. The class label of the selected object (e.g., "cup", "person") and the calculated dynamic scale factor are used to query a pre-defined shape database (detailed in Section 3.3.4). This database contains simplified drawing commands (G-code sequences) for various object classes. The pipeline retrieves the corresponding scaled G-code sequence from the database, which represents the tactile pattern to be drawn by the Line-us plotter.

This pipeline operates periodically on captured frames (controlled by a FRAME_PROCESS_INTERVAL) to balance responsiveness with computational load, ensuring that the system can process visual input and generate tactile commands in near real-time.

3.3.3. Line-us Communication Software

The software component responsible for interfacing with the Line-us plotter handles the wireless transmission of drawing commands. Communication with the Line-us device is established over a **TCP/IP socket**

connection, typically via Wi-Fi, leveraging the plotter's built-in network capabilities.

The Android application, running the Python-based logic, opens a socket connection to the Line-us's designated IP address or hostname and port (port 1337 is standard). Drawing instructions are sent as **G-code commands**, specifically using the G01 command format, which instructs the plotter to perform a linear move to a specified X, Y, and Z coordinate. The software constructs sequences of these G01 commands based on the scaled shape data retrieved from the database (as described in Section 3.3.2). The Z-coordinate within these commands is crucial for controlling the pen state, differentiating between movements with the pen head lifted (PEN_UP_Z) and movements with the pen head in contact with the skin (PEN_DOWN_Z).

To prevent the application's user interface from becoming unresponsive during the drawing process, the communication and command execution sequence with the Line-us is performed in a **separate background thread**. This allows the main application thread to continue processing camera frames and updating the display while the plotter is physically executing the drawing commands. The software also includes logic to manage the connection state, attempting to connect or reconnect to the Line-us as needed and handling potential communication errors. As discussed in Section 3.3.1, the implementation of this communication protocol was based on the available Line-us programming documentation. [22]

3.3.4. Shape Database

A fundamental element of the chosen object detection and database mapping approach is the **Shape Database**. Implemented as a Python module (shapes_db.py), this database serves as a repository for pre-defined,

simplified drawing sequences corresponding to specific object classes that the YOLOv3-tiny model is trained to detect.

Each entry in the database is associated with an object class label (e.g., "cup", "chair") and contains a sequence of G-code commands (G01 X Y Z) that define the movements required for the Line-us plotter to draw a simplified, abstract representation of that object. These drawing sequences are designed to be easily recognizable through tactile sensation on the skin.

Crucially, the database is structured to support **dynamic scaling**. When the system queries the database with a detected object's class label and a calculated scale factor (derived from the object's size in the camera frame, as discussed in Section 3.3.2), the function (get_gcode(shape_name, scale_factor) is used for this. the function returns the corresponding G-code sequence with all coordinates appropriately scaled. This allows the size of the tactile drawing to adapt to the perceived size or distance of the object in the visual input, enhancing the user's spatial understanding. The database centralizes the drawing logic for each recognizable object, providing a consistent and reliable source for generating tactile patterns.

3.3.5. Database for Storing Shapes' Scrips

For the separate thermal tactile display prototype, the software runs on the **ESP32 microcontroller**. This firmware is designed to operate the ESP32 as a **web host**, allowing a client device (such as the Android smartphone) to connect to it wirelessly. The software on the ESP32 contains a set of **pre-defined thermal patterns**. Upon receiving a command or signal (presumably from the connected client, although full integration with the object detection pipeline was not completed), the ESP32 selects the corresponding pre-defined pattern. This pattern is then translated into control

signals sent to the ULN drivers, which manage the activation of specific resistors in the 5x4 matrix to generate the desired thermal shape on the skin surface.

3.4. PARTICIPANTS

This section details the recruitment process and characteristics of the individuals who participated in the user study to evaluate the effectiveness of the plotter-based tactile stimulator system.

3.4.1. Recruitment

Participants for the study were recruited through a combination of outreach methods. This included inviting friends and family to volunteer, as well as conducting outreach to local organizations and individuals who might know potential participants, particularly those with visual impairments. The aim was to gather a diverse group willing to engage with the tactile feedback system.

Criteria for Selection:

1. **Age:** Participants were selected from various age groups to evaluate how different ages may affect the interpretation of tactile feedback.
2. **Sight Ability:** A mix of blind and sighted individuals was included to compare their experiences with the device.
3. **Willingness:** All participants expressed a willingness to engage in the study and provided informed consent.

Table 3.1 | Age group, gender, and vision state of the 20 participants.

Age	8	10	11	13	15	18	19	21	21	23	25	25	26	29	31	33	37	41	59	61
Gender	M	F	M	M	M	M	M	M	M	M	F	M	M	F	F	F	M	M	F	M
State	G	B	G	L	L	G	G	L	G	G	L	G	G	G	L	G	G	L	L	L
Educated	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	Y	N	N
Note	**M:** Male **F:** Female **G:** Good Vision **L:** Low Vision **B:** Blind **Y/N:** Yes/No																			

3.4.2. Participant Demographics

A total of **20 individuals** participated in the user study. This cohort constituted a **single group** with mixed demographics, including varying age groups and different vision statuses (sighted, low vision, and blind individuals). The diversity in age and vision aimed to explore the system's potential applicability and learning characteristics across a broader population. The specific age distribution, gender mix, and vision status of the participants are summarized in **Table 3.1** above.

3.4.3. Ethical Considerations

Throughout the recruitment and experimental phases, strict ethical considerations were observed. All participants were provided with comprehensive information regarding the study's purpose, procedures, potential discomforts, and benefits. **Informed consent** was obtained from each participant before their involvement. Participant identities were kept **confidential**, with data being anonymized for analysis and reporting to protect their privacy. Participants were also informed of their **right to withdraw** from the study at any point without facing any negative consequences.

3.5. EXPERIMENTAL PROCEDURE

This section details the experimental procedures followed to evaluate the effectiveness of the tactile stimulator system. The evaluation process included an initial manual phase conducted in the first semester to explore the basic feasibility of tactile shape recognition through drawing on the skin and to investigate factors such as **skin area sensitivity** and suitable drawing tools. This was followed by a comprehensive user study in the second semester utilizing the developed plotter-based system.

The **initial manual phase** involved participants attempting to recognize simple shapes drawn directly onto their skin by a facilitator. This preliminary testing, conducted without the automated device, served as a proof-of-concept to assess the human sensory system's capacity to interpret drawn tactile patterns and to identify suitable drawing tools and skin areas based on tactile feedback and sensitivity variations. The insights gained from this manual exploration informed the subsequent design and experimental protocols for the automated system, including the selection of the forearm/hand as the primary drawing location.

Following this initial phase, a structured user study was conducted with the implemented plotter-based tactile stimulator, as described in the following subsections.

3.5.1. Training Phase

The user study commenced with a structured training phase designed to familiarize participants with the plotter-based tactile stimulator and enable them to effectively interpret the tactile patterns drawn on their skin. This phase consisted of **six dedicated training sessions**.

During these sessions, participants were introduced to the wearable Line-us device and its operation. The training focused on recognizing a set of **shapes representing real-world objects**, which were drawn on their skin by the plotter using the selected feather brush pen head. These shapes were sourced from the system's pre-defined shape database (as described in Section 3.3.4).

Participants practiced identifying these shapes through tactile sensation. In the initial training sessions, when a participant misidentified a shape, the correct shape was revealed and, if necessary, redrawn to reinforce the association between the tactile pattern and the object it represented. This corrective feedback in the early stages aimed to accelerate the learning process and build participant confidence in interpreting the tactile stimuli. The training sessions provided repeated exposure to the tactile patterns, allowing participants to develop and refine their ability to discriminate and recognize the shapes solely through touch.

3.5.2. Testing Phase

Following the training phase, participants proceeded to the testing phase to formally evaluate their tactile shape recognition abilities using the developed plotter system. This phase consisted of **ten testing sessions** conducted over a period of time to observe learning progression.

In each testing session, participants were presented with a series of shapes drawn by the Line-us plotter on their skin. These shapes included those they had been trained on, representing various real-world objects from the shape database. Participants were asked to identify each shape solely based on the tactile feedback received. To assess their learning and

adaptability, **new, previously unseen shapes** were also introduced in later testing sessions.

Accuracy was the primary metric recorded during this phase. For each shape presented, the participant's guess was noted as either correct or incorrect. Unlike the initial training sessions, corrective feedback was not provided during the formal testing trials to obtain an unbiased measure of recognition performance at that point in time. The data collected from these sessions formed the basis for the quantitative analysis of the system's effectiveness and user learning curves.

3.5.3. Data Collection and Analysis

Data from the testing phase was systematically collected to quantify participant performance and assess the effectiveness of the tactile stimulator system. During each testing session, the participant's identification of each presented shape was recorded. This data primarily consisted of noting whether each guess was correct or incorrect for every shape presented in a session.

The collected data was primarily managed and analyzed using **Google Sheets**. This platform facilitated the recording of results across all participants and sessions and allowed for automated calculation of key performance metrics.

The analysis focused on evaluating participant learning and system efficacy over the ten testing sessions. Several metrics were calculated based on the recorded correct guesses:

1. **Accuracy of Shape Recognition:** Calculated as the percentage of correctly identified shapes out of the total number of shapes presented

in a given session for each participant and averaged across groups or the entire cohort.

2. **Learning Speed and Improvement:** Assessed by tracking the change in accuracy over the testing sessions. Specific calculations included:
   - **Improvement:** The difference in average accuracy between the final testing session and the first testing session.
   - **Percentage Improvement:** The Improvement value expressed as a percentage relative to the average accuracy on the first testing session.
   - **Over-Sessions Rate of Change:** The change in average accuracy from one testing session to the next.
   - **Average Over-Sessions Rate of Change:** The average of the Over-Sessions rates of change across all testing intervals.

The formulas used for calculating Improvement, Percentage Improvement, Over-Sessions Rate of Change, and Average Over-Sessions Rate of Change were as follows:

Equation 3.1 |

$$\boldsymbol{Improvment} = \boldsymbol{Average\ on\ Session\ (Last)} - \boldsymbol{Average\ on\ Session\ (First)}$$

Equation 3.2 |

$$\boldsymbol{Percentage\ Improvment} = \frac{\boldsymbol{Improvment}}{\boldsymbol{Average\ on\ Session\ (First)}} \times \mathbf{100\%}$$

Equation 3.3 |

$$\boldsymbol{OverSessions\ Rate\ of\ Change\ (Session\ X)} = \boldsymbol{Average\ on\ Session\ (X+1)} - \boldsymbol{Average\ on\ Session\ X}$$

Equation 3.4 |

$$Average\ OverSessions\ Rate\ of\ Change\ = \frac{Sum\ of\ OverSession\ Changes}{Number\ of\ Changes}$$

This analytical approach provided quantitative insights into how well participants learned to interpret the tactile feedback, the rate of their learning, and the overall performance of the system in facilitating tactile shape recognition.

4. RESULTS

4.1. OVERVIEW OF TESTING OUTCOMES

The user study involved 20 participants of mixed vision status and age groups, who underwent a structured training and testing protocol using the developed tactile stimulator. The evaluation focused on their ability to recognize a set of shapes representing real-world objects drawn on their skin by the Line-us plotter. Data collected across ten testing sessions provided insights into the system's effectiveness in facilitating tactile learning and shape discrimination.

4.2. DATA PRESENTATION

4.2.1. Accuracy of Shape Recognition

Participant accuracy in recognizing tactile shapes showed a general trend of improvement over the testing period. The overall average accuracy across all 20 participants increased from an initial percentage in the first testing session to a higher percentage by the final session. This learning curve indicates that participants were able to adapt to and learn from the tactile feedback provided by the plotter system. Table **4.1** summarizes the average percentage of correct shape recognition across all participants for each of the ten testing Sessions.

Table 4.1 | The accuracy of shape for 20 Participants

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
P1	42.11%	52.63%	52.63%	52.63%	57.89%	52.63%	57.89%	42.11%	73.68%	89.47%
P2	23.08%	53.85%	53.85%	53.85%	61.54%	53.85%	69.23%	69.23%	76.92%	92.31%
P3	57.89%	73.68%	63.16%	63.16%	78.95%	63.16%	73.68%	78.95%	68.42%	94.74%
P4	36.84%	47.37%	52.63%	47.37%	57.89%	47.37%	63.16%	52.63%	57.89%	84.21%
P5	42.11%	52.63%	42.11%	52.63%	57.89%	52.63%	57.89%	47.37%	52.63%	89.47%
P6	52.63%	63.16%	68.42%	57.89%	73.68%	57.89%	63.16%	73.68%	73.68%	78.95%
P7	21.05%	42.11%	42.11%	47.37%	52.63%	57.89%	63.16%	68.42%	63.16%	68.42%
P8	57.89%	68.42%	52.63%	63.16%	84.21%	63.16%	73.68%	84.21%	84.21%	89.47%
P9	42.11%	63.16%	52.63%	57.89%	52.63%	73.68%	68.42%	68.42%	73.68%	84.21%
P10	31.58%	42.11%	52.63%	57.89%	57.89%	68.42%	68.42%	52.63%	57.89%	84.21%
P11	42.11%	47.37%	52.63%	47.37%	63.16%	73.68%	78.95%	68.42%	84.21%	94.74%
P12	42.11%	47.37%	57.89%	47.37%	68.42%	63.16%	63.16%	52.63%	68.42%	84.21%
P13	42.11%	47.37%	57.89%	47.37%	68.42%	63.16%	63.16%	52.63%	68.42%	84.21%
P14	31.58%	42.11%	47.37%	47.37%	52.63%	47.37%	63.16%	63.16%	68.42%	63.16%
P15	47.37%	42.11%	57.89%	52.63%	57.89%	52.63%	63.16%	68.42%	78.95%	68.42%
P16	42.11%	52.63%	63.16%	68.42%	52.63%	57.89%	68.42%	52.63%	47.37%	63.16%
P17	31.58%	47.37%	47.37%	47.37%	47.37%	42.11%	52.63%	47.37%	52.63%	57.89%
P18	26.32%	42.11%	42.11%	47.37%	42.11%	47.37%	57.89%	52.63%	47.37%	68.42%
P19	26.32%	36.84%	36.84%	36.84%	42.11%	36.84%	42.11%	42.11%	47.37%	57.89%
P20	21.05%	36.84%	42.11%	42.11%	47.37%	42.11%	52.63%	47.37%	47.37%	63.16%
S: Session **P**: Participant										

The overall learning progression is further illustrated by the learning curve, which plots the average recognition accuracy over the testing sessions.

Table 4.2 summarizes the average of the accuracies for different age groups, and the overall accuracy over all sessions.

Table 4.2 | The overall percentage of accuracy for all participants as well as for different age groups

LA: Learning Accuracy											Groups
LA*	40.40%	56.03%	52.87%	53.93%	62.83%	53.93%	64.37%	58.06%	65.91%	90.04%	7-15 Years
LA	40.35%	51.46%	53.80%	52.63%	63.74%	63.16%	67.25%	64.91%	71.35%	81.29%	16-29 Years
LA	32.46%	42.98%	48.25%	49.12%	48.25%	46.49%	56.14%	51.75%	53.51%	63.16%	30-65 Years
LA	38.00%	50.06%	51.90%	51.90%	58.87%	55.85%	63.20%	59.25%	64.64%	78.04%	Overall

4.2.2. Learning Speed and Improvement:

Beyond overall accuracy, the study also evaluated the speed and extent of participant learning throughout the testing phase. Analysis of the data revealed a notable improvement in shape recognition performance from the initial sessions to the final ones. (See below **Figure 4.1**)

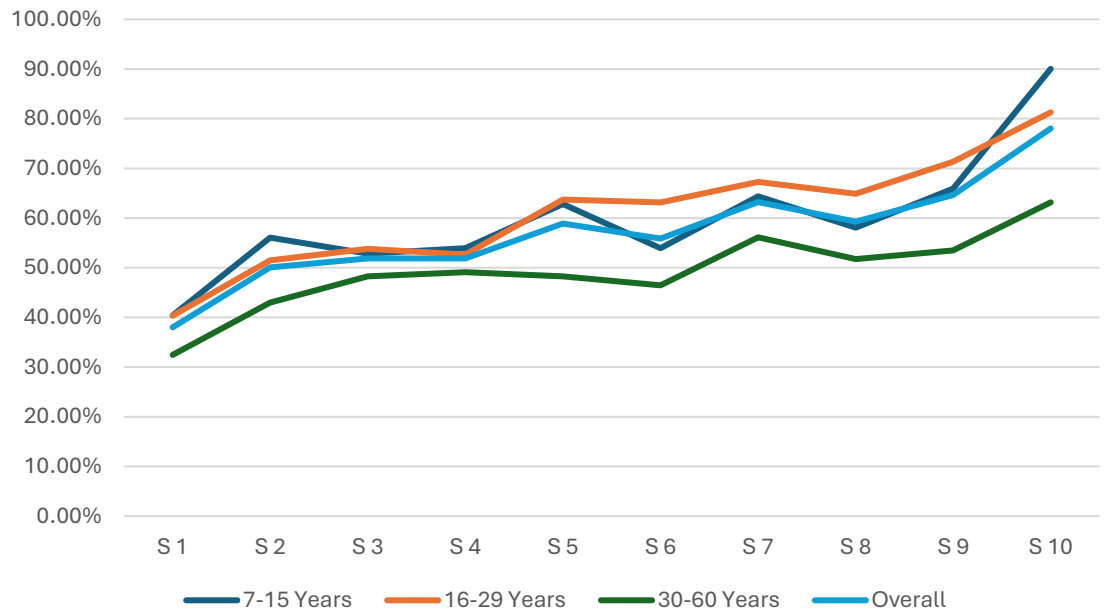


Figure 4.1 | The Learning curves for different age groups and for overall participants.

For the entire cohort of 20 participants, the overall **Improvement** in average accuracy from the first testing session to the last was calculated to be 40.04 percentage points. Correspondingly, the **Percentage Improvement** over the testing period was 105.38%. These figures indicate a significant positive change in participants' ability to interpret the tactile feedback over time.

The **Average Over-Sessions Rate of Change** in accuracy across all participants was 3.73% percentage points per session, suggesting a consistent, albeit variable, learning rate throughout the study.

Further analysis revealed differences in learning patterns across various age groups within the mixed participant cohort. While all groups showed improvement, the rate and magnitude of learning varied.

Table 4.3 | Improvement, Percentage Improvement and Average Over-Sessions Rate of Changes for the age groups, and the overall.

Results	overall	0-11	12-29	30-65
Improvement	40.04%	49.64%	40.94%	30.70%
Percentage Improvement	105.38%	122.85%	101.45%	94.59%
Average Over-Sessions Rate of Change	3.73%	5.31%	3.93%	2.13%
Over-Sessions Rate of Change (S X)				
S X-Y	overall	0-11	12-29	30-65
S 1-2	12.06%	15.63%	11.11%	10.53%
S 2-3	1.84%	-3.16%	2.34%	5.26%
S 3-4	0.00%	1.05%	-1.17%	0.88%
S 4-5	6.96%	8.91%	11.11%	-0.88%
S 5-6	-3.02%	-8.91%	-0.58%	-1.75%
S 6-7	7.35%	10.45%	4.09%	9.65%
S 7-8	-3.95%	-6.32%	-2.34%	-4.39%
S 8-9	5.38%	7.85%	6.43%	1.75%
S 9-10	13.40%	24.13%	9.94%	9.65%

Figure 4.2 illustrates the learning Improvement for different age groups (e.g., 0-11 years, 12-29 years, 30-60 years), showing how far they have improved during the usage of system. It tells us that children have more capacity to adapt with the system usage than other age groups.

Figure 4.2 | The Figure shows that the children are more adaptive to the system than other age groups.

4.2.3. Impact of Shape Complexity

The study also investigated how the complexity and characteristics of the individual shapes affected participants' ability to recognize them through tactile feedback. Analysis of the accuracy rates for each shape revealed variations in recognition performance.

As shown in Table 4.4, simpler geometric shapes, such as **Triangle** or the **Undefined** shape which is the fallback shape when an undefined object is detected, represented by a straight line, generally exhibited higher recognition accuracy compared to more complex shapes representing real-world objects, like **Cat**. This suggests that the level of detail and the number of distinct features in a tactile pattern influence the ease with which it can be interpreted by touch. Shapes with fewer points or simpler outlines were more consistently and accurately identified by participants.

Table 4.4 | The average recognition accuracy for each shape tested across all participants over the ten testing sessions.

Shape	Sum
?	96.32%
Circle	86.84%
Triangle	87.89%
Square	83.16%
person	56.32%
bicycle	55.79%
car	39.47%
cup	64.74%
cat	23.68%
chair	32.63%
book	43.16%
bottle	51.58%
cell phone	44.74%
apple	37.37%
umbrella	68.42%
tvmonitor	35.79%
handbag	57.37%
banana	47.89%
fork	60.53%

4.3. COMPARATIVE ANALYSIS

In addition to the quantitative performance metrics, valuable qualitative data was gathered through subjective feedback from participants after the testing sessions. Participants were encouraged to share their experiences regarding the usability, comfort, and perceived effectiveness of the tactile stimulator system.

The feedback provided insights into various aspects of the system, particularly concerning the tactile interface. Regarding the pen heads explored (as discussed in Section 3.2.2), participants expressed preferences based on the resulting sensation. While the feather brush was chosen for the main study, comments highlighted the different tactile qualities perceived with other tools tested during the initial phase. For instance, one participant noted, **"I can feel the shapes better when you're doing with the brush,"** supporting its selection. Another interesting observation from the initial manual testing phase was the clarity of sensation when using a pen with a thin material over the skin, as one participant described, **"when you put the plastic bag on my hand and moved the pen over it, I was feeling everything, just like I was actually seeing!"**

Feedback also underscored the previously noted variations in skin sensitivity across different body areas (as explored in the initial manual phase). Participants reported differences in how clearly they perceived the tactile patterns depending on where they were drawn. One participant commented on the distinct sensation on a specific area, stating, **"The pen feels ticklish when you move it over my palm,"** while another observed, **"I feel I know more details when you draw on my palm than the back of my hand."** These comments reinforce the importance of considering skin sensitivity when designing tactile feedback systems and selecting the optimal placement area.

Comments related to the wearable design, comfort during use, and the overall experience of interacting with the system were also recorded. This subjective feedback complements the quantitative results, providing a richer understanding of the user experience and highlighting areas of success and potential improvement for the tactile stimulation approach.

5. DISCUSSION

5.1. INTERPRETATION OF RESULTS

The quantitative results from the user study provide compelling evidence that participants were able to learn and improve their tactile shape recognition skills using the developed plotter-based system. The most prominent finding is the significant increase in overall average accuracy across the ten testing sessions, rising from an initial 38.00% in Session 1 to 78.04% in Session 10 (**Table 4.2**). This substantial improvement, quantified as a 40.04 percentage point increase and a 105.38% percentage improvement (**Table 4.3**), clearly demonstrates the human sensory system's capacity to adapt to and interpret dynamic tactile patterns drawn on the skin. The consistent positive Average Over-Sessions Rate of Change (3.73%) further underscores that learning was an ongoing process throughout the study period.

Analysis of the data segmented by age group revealed interesting differences in learning trajectories. While all age categories showed improvement, the youngest group (7-15 years) exhibited the highest percentage improvement (122.85%) and the highest average over-sessions rate of change (5.31%) (**Table 4.3**). This suggests that younger individuals may possess greater neuroplasticity or adaptability, allowing them to more rapidly acquire the skills needed to interpret this novel form of tactile feedback. The older age groups also demonstrated significant learning,

although at a somewhat slower rate compared to the youngest participants (**Table 4.3**, **Figure 4.1** and **Figure 4.2**).

The study also confirmed that the characteristics of the tactile patterns themselves play a crucial role in recognition accuracy. Simpler shapes with fewer lines or more distinct features, such as the '?' or Triangle, were consistently recognized with higher accuracy (96.32% and 87.89% respectively) compared to more complex shapes like 'cat' (23.68%) (**Table 4.4**). This highlights that while the system can deliver various patterns, the inherent complexity of the shape directly impacts the cognitive load and discriminability for tactile interpretation. Simplifying complex visual information into optimally discernible tactile patterns is a key consideration for future development.

These quantitative findings are further supported by the subjective feedback gathered from participants, which provided qualitative insights into their perceptual experiences and preferences regarding the tactile interface and drawing characteristics.

5.2. COMPARISON OF APPROACHES

The development of the tactile stimulator involved evaluating multiple strategies for converting visual information into tactile feedback. As detailed in Section 3.1.3, three primary automated approaches were explored for the plotter system: image processing and contour grouping, object detection with ROI contour processing, and object detection with shape database mapping. Based on practical implementation and initial testing, the **object detection with shape database mapping** approach was found to be the most effective for this project. This method demonstrated superior **speed, robustness to environmental clutter, and consistency** in generating tactile patterns

compared to the contour-based methods, which struggled with noise, lighting variations, and reliably isolating complex shapes from live camera feeds.

Beyond the different automated processing pipelines, the project also provided an opportunity to compare the developed plotter-based system against alternative methods of delivering tactile feedback explored during the research. The **initial manual drawing phase**, conducted in the first semester, served as a foundational exploration into the feasibility of tactile shape recognition through drawing on the skin. This manual phase allowed for direct investigation into factors like the effectiveness of different **drawing tools** (leading to the selection of the feather brush for the automated system, as discussed in Section 3.2.2) and the varying **skin sensitivity** across different body areas. While valuable for proof-of-concept and parameter selection, manual drawing lacks the real-time responsiveness and consistency required for a dynamic sensory substitution device. The automated plotter system successfully translated this manual concept into a consistent, repeatable, and real-time process.

Furthermore, a **thermal tactile display prototype** (Section 3.2.6 and 3.3.5) was developed as a separate exploration into conveying information through temperature changes. While this prototype demonstrated the feasibility of generating thermal patterns on the skin, preliminary testing indicated that the resolution and clarity of shapes conveyed through the 5x4 resistor matrix were **limited** compared to the continuous lines drawn by the plotter. The plotter's ability to draw arbitrary shapes with a resolution limited only by the pen head size and skin sensitivity appeared to offer a richer and more discernible tactile experience for shape recognition in initial comparisons. This suggests that, for the purpose of conveying distinct

shapes, the plotting-on-the-skin approach holds significant advantages over the tested thermal matrix method.

5.3. IMPLICATIONS FOR SENSORY SUBSTITUTION

The findings from this study have significant implications for the field of tactile sensory substitution and the development of assistive technologies. Existing sensory substitution devices, such as electro-tactile arrays (e.g., BrainPort)[5] and vibrotactile systems, often face several limitations that can hinder their widespread adoption and effectiveness. These challenges commonly include **high manufacturing costs**, **steep learning curves requiring extensive user training and adaptation periods**, **limited spatial resolution** in tactile displays which can restrict the detail of conveyed information, and issues related to the **comfort and wearability** of bulky or invasive devices.

The developed plotter-based tactile stimulator offers a novel approach that potentially addresses some of these limitations. By utilizing a modified, relatively **low-cost off-the-shelf plotter** and standard mobile computing hardware, the system presents a more economically accessible alternative compared to many specialized sensory substitution devices. Furthermore, the method of creating **continuous tactile drawings directly on the skin** provides a different form of tactile feedback compared to discrete point-based or vibrotactile arrays. This continuous nature, which offers a form of tactile resolution akin to drawing, may provide a more intuitive and potentially richer spatial understanding for users, contributing to a less steep learning curve for recognizing shapes. The wearable design, focusing on placement on the forearm or hand, aims to improve comfort and practicality for everyday use.

While this study focused on shape recognition, the ability of the system to translate visual objects into discernible tactile patterns holds promise for enhancing spatial awareness and object recognition for visually impaired individuals. The system's architecture, based on readily available technology and an open-source framework (Kivy, OpenCV), also contributes to the potential for further development and accessibility within the research community. Beyond assisting the visually impaired, this approach could have applications in areas requiring non-visual information display, such as immersive virtual reality environments or specialized training simulations.

5.4. LIMITATIONS AND CHALLENGES

While the developed plotter-based tactile stimulator system demonstrated promising results in facilitating tactile shape recognition, several limitations inherent to the study design and the current prototype should be acknowledged. Addressing these challenges will be crucial for the future development and practical application of the system.

One significant limitation of the user study is the **sample size and composition**. While the inclusion of participants with mixed vision statuses provided valuable initial insights into the system's potential applicability across different user groups, the total number of 20 participants is relatively small. A larger, more diverse cohort, potentially with a greater focus on individuals with significant visual impairments, would be necessary to draw more generalized conclusions about the system's effectiveness and to investigate differences in learning and performance based on the degree and onset of vision loss.

Furthermore, the **testing environment** was primarily controlled. Evaluating the system's performance in dynamic, real-world environments

presents additional challenges, such as varying lighting conditions, complex backgrounds, and the presence of multiple moving objects. While the object detection approach offers improved robustness compared to contour-based methods, accurately identifying and tracking objects in highly cluttered or rapidly changing scenes can still be difficult. The current system's focus on selecting a single central object also limits its ability to convey information about the spatial relationships between multiple objects present in the environment simultaneously.

The **specific set of shapes** included in the shape database also represents a limitation. While the study demonstrated learning with a set of shapes representing real-world objects, the variety and complexity of objects encountered in everyday environments are vast. Expanding the shape database and developing strategies for handling objects not present in the database are necessary steps for increasing the system's practical utility. As shown in the results (Section 4.2.3), the complexity of the shape significantly impacts recognition accuracy, highlighting the challenge of simplifying complex visual forms into discernible tactile patterns.

Challenges were also encountered related to the **wearable design and user comfort**. While efforts were made to make the Line-us portable with a battery and select a comfortable pen head, ensuring comfort during prolonged use and developing a mounting mechanism that remains stable across various user activities are ongoing considerations. Potential issues such as skin irritation or the physical bulk of the device could affect user acceptance and adherence.

Finally, the **thermal tactile display prototype**, while explored as an alternative modality, demonstrated limitations in its current form, particularly regarding the resolution and clarity of the thermal patterns

compared to the plotter's line drawings (5.2). Further research is needed to determine if thermal feedback can effectively convey complex spatial information.

5.5. FUTURE DIRECTIONS

The successful design, implementation, and evaluation of the plotter-based tactile stimulator prototype open several promising avenues for future research and development. Building upon the foundation established in this project, future work could focus on enhancing the system's capabilities, improving the user experience, and exploring advanced hardware implementations.

One critical area for future work involves **advancing the visual processing and tactile rendering pipeline**. To overcome the limitation of being restricted to a pre-defined shape database, future iterations could explore the integration of **artificial intelligence (AI) or machine learning algorithms** capable of **generating simplified drawing patterns in real-time** directly from the detected object's visual features, even for objects not explicitly included in a database. This would significantly expand the range of objects the system can represent and enhance its adaptability to novel environments. Further research is also needed to develop strategies for handling **multiple detected objects** simultaneously and conveying their spatial relationships through tactile feedback.

Enhancements to the **wearable design** are also essential. Future iterations should focus on developing lighter, more comfortable, and more stable mounting mechanisms that can be worn for extended periods across various activities. Investigating different skin contact materials or techniques

could further improve tactile clarity and user comfort, potentially mitigating issues like ticklishness observed in some areas.

Looking further ahead, the potential for **device miniaturization** is a significant future direction. While the current prototype utilizes a modified off-the-shelf plotter, the ultimate goal could be the development of a much smaller, purpose-built tactile rendering device. The advent of **Micro-Electro-Mechanical Systems (MEMS)**[20], [21] technology offers a promising path towards creating highly miniaturized tactile actuators. This could enable the development of a device comparable in size and wearability to a watch, incorporating a small, precise moving part capable of drawing continuous tactile patterns directly onto the skin of the wrist or forearm, offering a discreet and highly integrated sensory substitution experience.

To gain a deeper understanding of the system's impact, **more extensive user studies** are recommended. This includes conducting studies with larger and more specifically defined cohorts, particularly focusing on individuals with varying degrees and types of visual impairments, to assess long-term learning, adaptation, and the practical utility of the device in daily life. Investigating the effectiveness of different training protocols could also optimize the learning process.

Exploring **enhanced accessibility features** is crucial for making the system more usable for visually impaired individuals. Developing and integrating a dedicated hardware control piece, such as a tactile or audio-based remote, could provide a non-visual means for users to interact with the system, select objects, or adjust settings.

6. CONCLUSION

This project successfully designed, implemented, and evaluated a wireless wearable tactile stimulator that utilizes a Line-us robotic plotter to create dynamic tactile patterns on the skin based on visual input. The core system, controlled by an Android application employing object detection and a shape database, demonstrated the feasibility of translating environmental information into a tactile format. The user study involving 20 participants showed significant learning and improvement in tactile shape recognition over ten testing sessions, with overall average accuracy increasing substantially and a notable percentage improvement observed across the cohort. These results indicate the human sensory system's capacity to adapt to and interpret continuous tactile feedback delivered through plotting on the skin, highlighting the potential of this novel approach for conveying spatial information.

The developed system offers a promising, potentially lower-cost alternative to some traditional sensory substitution devices by leveraging readily available technology and providing a form of continuous tactile feedback that may enhance intuitive understanding. While the current prototype has limitations, including reliance on a pre-defined shape database and challenges related to wearability and real-world performance, it lays a strong foundation for future advancements. Exciting future directions include the integration of artificial intelligence for real-time generation of

tactile patterns for any detected object, overcoming the database constraint, and the long-term vision of miniaturizing the technology using advancements like MEMS for highly integrated wearable devices. Ultimately, this work contributes to the field of assistive technology by exploring a new method for tactile sensory substitution, opening possibilities for enhancing independence and interaction with the environment for individuals with sensory impairments.

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APPENDICES

- **Appendix I:** Source Codes (1) And Details (2) Of the Three Approaches for All the Iterative Process. Link (1): https://drive.google.com/drive/folders/1ECdDnp_jiz6tC45jYaYhgiSx7Qu9d_gQ?usp=drive_link
  Link (2): https://drive.google.com/drive/folders/1KK9ZpddoExlsB-Lfyq8EvEYhP5hhbkgw?usp=drive_link

- **Appendix II:** Source Code of The Android Application with the APK App. Link: https://drive.google.com/drive/folders/1J7PKty_aYDplSlqt8xmBr0OPneqYa3SW?usp=drive_link

- **Appendix III:** Experiment Row Data. Link: https://drive.google.com/drive/folders/1LB-ACL-mm0LuR3meRO5Vnh0kKkP1yBrw?usp=drive_link

- **Appendix IV:** Object Used. Link: https://drive.google.com/drive/folders/1CWw2HEv0I380Ddzh185_UqNsNlhiJnyH?usp=drive_link

- **Appendix V:** ESP32 Programming Code. Link: https://drive.google.com/drive/folders/1Lib1wwUlZVqqvqTf_AsfvrQOEwrfH7BA?usp=drive_link

- **Appendix VI:** GitHub Repository to preview the markdown files. Link: https://github.com/shakerbr/FinalYearProject-SB.git

الملخص

تقدم هذه الدراسة تصميم وتنفيذ محفز لمسي مبتكر يهدف إلى تعزيز النظام الحسي البشري باستخدام أسلوب التخطيط على الجلد. لمعالجة القيود في أجهزة استبدال الحواس الحالية، مثل التكلفة العالية، ودقة العرض المحدودة، ومتطلبات التدريب المكثفة، طور هذا المشروع نظامًا لاسلكيًا يمكن ارتداؤه يتمحور حول جهاز رسم آلي Line-us معدّل. يستخدم النظام تطبيق أندرويد مخصصًا يعالج المدخلات المرئية من الكاميرا باستخدام تقنية اكتشاف الأجسام YOLOv3-tinyوقاعدة بيانات للأشكال لتوليد أوامر رسم مبسطة. يتم إرسال هذه الأوامر لاسلكيًا إلى جهاز الرسم Line-usالمعدّل ببطارية، والذي يستخدم فرشاة ريشة لرسم أنماط لمسية مباشرة على جلد المستخدم. تم إجراء دراسة مستخدم بمشاركة 20 فردًا من حالات بصرية مختلفة وفئات عمرية متنوعة على مدار عشر جلسات اختبار لتقييم التعرف على الأشكال باللمس. أظهرت النتائج تعلمًا وتحسنًا كبيرًا في دقة التعرف على الأشكال بين المشاركين، مما يشير إلى قدرة الجهاز الحسي البشري على التكيف وتفسير ردود الفعل اللمسية المستمرة المقدمة عن طريق التخطيط. كما سلطت النتائج الضوء على تأثير تعقيد الشكل على أداء التعرف وقدمت ملاحظات ذاتية قيمة حول الواجهة اللمسية. يساهم هذا العمل في تقديم إثبات مفهوم لطريقة استبدال حسي لمسي يحتمل أن تكون أقل تكلفة وأكثر سهولةً في الاستخدام، وتوفر شكلاً من ردود الفعل اللمسية المستمرة. تشمل الاتجاهات المستقبلية الاستفادة من الذكاء الاصطناعي لتوليد الأشكال في الوقت الفعلي للتغلب على قيود قاعدة البيانات واستكشاف إمكانيات التصغير من خلال التطورات مثل تقنية الأنظمة الكهروميكانيكية الدقيقة.(MEMS)

جمهورية العراق
وزارة التعليم العالي والبحث العلمي
جامعة الموصل
قسم هندسة الحاسوب

تصميم وتنفيذ محفز لمس لتعزيز الجهاز الحسي البشري باستخدام أسلوب التخطيط على الجلد

مشروع تخرج مقدم الى قسم هندسة الحاسوب - جامعة الموصل كجزء من متطلبات نيل شهادة البكالوريوس في كلية الهندسة قسم هندسة الحاسوب

من قبل

شاكر بشار حسين

المشرف

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موصل، عراق

2024-2025